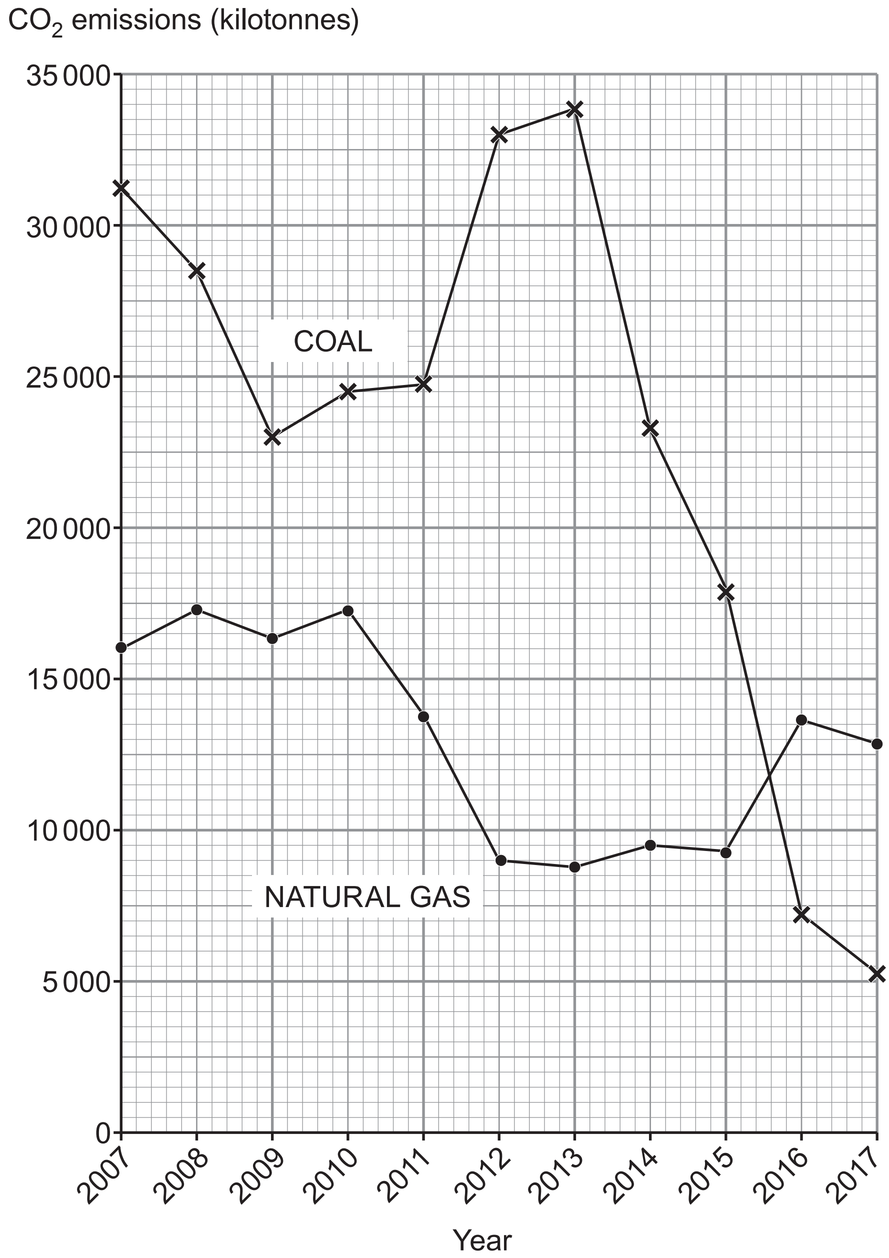
Examiner only

Answer **all** questions.

1. The UK government wants to reduce the CO_2 emissions from power stations.

The graphs below show the CO_2 emissions from coal and natural gas power stations between the years 2007 and 2017.



Examiner only

(a) Look at the data for **2012**. Calculate the difference in the emissions of CO_2 from coal and natural gas. [2]

difference = ... kilotonnes of CO_2

(b) Between the years 2014 and 2015, the emission of CO_2 from coal fell by 5500 kilotonnes.

State between which other years the emission of CO_2 from coal fell **at the same rate**.[1]

Years .... and ....

(c) State **two** benefits of reducing CO_2 in the atmosphere. [2]

1. ....

2. ....

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(d) (i) Nuclear power stations provide up to 20% of the present UK demand for electricity.

Gas provides up to 50%.

One student, Seren, says that a graph for the CO_2 emissions from nuclear power stations would be the same shape as for gas but always lower.

Explain whether you agree with Seren. [2]

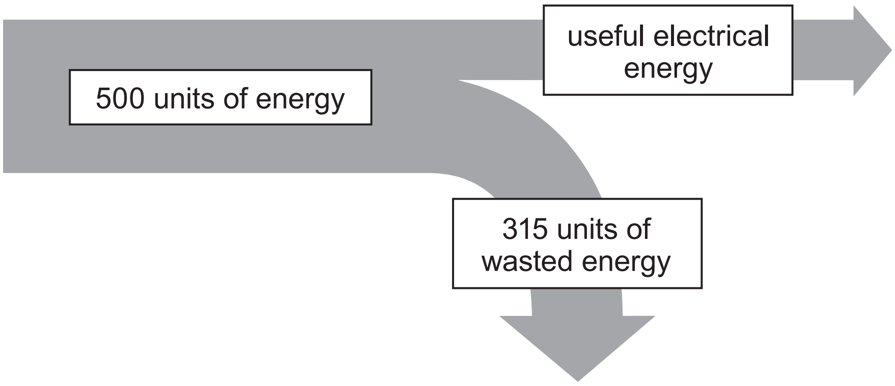
....

....

....

Examiner only

(ii) The Sankey diagram shows the energy input and output for a power station.



Seren looks at the diagram and calculates that the power station is 63% efficient.

Explain whether you agree with Seren. [2]

Space for calculation.

....

....

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